Nandan Illur

Embedded Systems Engineer — Embedded Linux — Hardware-Software Integration

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Education

KLE Technological University **Jan. 2022 – June 2025**

Bachelor of Science in Electronics and Communications *Hubballi, Karnataka*

CGPA: 7.4

Technical Skills

Languages: C, C++, Python, Shell Scripting

Embedded Systems: Embedded Linux, Raspberry Pi, Raspberry Pi CM4, ESP32, Arduino, ARM-based Systems, Board Bring-up

Firmware & Interfaces: UART, SPI, I^2C, GPIO, GPS/GNSS, GSM/4G, CAN

FPGA & Digital: Verilog, Digital Logic, RTL Fundamentals

Hardware & Data Acquisition: Camera Systems, Frame Grabbers, IMU, MPU6050, NEO-6M GPS

AI & Computer Vision: PyTorch, YOLOv8, Edge AI, Computer Vision

Tools: Git, GDB, VS Code, Proteus, LTspice

Experience

NeoCambrian AI **Sept. 2026 – Oct. 2026**

Embedded Systems Engineer – Contract *India*

- Performed board bring-up, system image building, flashing, configuration, and validation across multiple embedded platforms.
- Integrated processing boards with frame-grabber and camera systems for high-speed image acquisition, recording, and end-to-end data-path validation.
- Worked on hardware-software integration across multiple boards and peripherals, debugging boot, interface, communication, and data-transfer issues.
- Worked with embedded Linux-based systems for system configuration, deployment, and system-level validation.
- Debugged end-to-end embedded data pipelines involving compute platforms, camera interfaces, and peripheral hardware.

EiNETCORP **Jan. 2025 – June 2025**

AI Embedded Engineer Intern *Hubballi, Karnataka*

- Developed an assistive smart-glasses system integrating a camera, Raspberry Pi CM4, GPS, and 4G/GSM communication for visually impaired users.
- Integrated the SIM7600G-H 4G module with Raspberry Pi CM4 for real-time location tracking and communication.
- Developed and integrated physical input controls for image capture and video-related functions.
- Integrated sensors and peripherals into an embedded edge-AI wearable platform, addressing communication, power, and real-time processing requirements.

Projects

Multi-Camera Synchronized Vision System – System Architecture

- Designed a 4-camera synchronized vision architecture targeting 1920×1200 resolution at 60 FPS.
- Designed a GMSL2-based camera architecture using camera-side serializers, coaxial links, and host-side deserializers for synchronized multi-camera acquisition.
- Evaluated Jetson AGX Orin and Jetson Orin NX platforms considering compute, power, bandwidth, and system requirements.
- Designed a hardware synchronization strategy using FSYNC and considered camera calibration, bandwidth, storage, and system integration.

Sensor Fusion-Based EGO Localization Using Kalman Filter | *Python, Raspberry Pi* **Feb. 2024**

- Developed a sensor-fusion system on Raspberry Pi using MPU6050 and NEO-6M sensors, capturing and storing sensor data for processing.
- Applied a Kalman filter to fused sensor data to improve localization accuracy and demonstrate embedded sensor integration.

Real-Time Vehicle Crash Detection & Alert System | *Python, ESP32* **Oct. 2023**

- Integrated IMU, GPS, vibration, and ultrasonic sensors for accident detection and vehicle monitoring.
- Developed a real-time alert mechanism to transmit vehicle location and emergency notifications through communication modules.

Effect of SNR Degradation on DNN Object Detection | *YOLOv8, Python* Certificate **Dec. 2024**

- Generated annotated datasets with controlled SNR degradation and trained YOLOv8 models for object detection under varying noise conditions.
- Evaluated detection robustness using confidence scores, Intersection over Union (IoU), and mean Average Precision (mAP).

Certifications & Achievements

ISRO START Program **Sept. 2023**
Grade A – Space Technology and Applications *Certificate*

Accepted for Publication – Springer ICMISC 2024 **Mar. 2024**
GUI-based Multi-Model Approach for Alzheimer's Detection and Classification *Certificate*

Leadership

KLE Higher Studies Club **Jan. 2023 – Present**
Secretary *KLE Technological University*

- Led and coordinated a 30+ member student chapter and organized university outreach activities.